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Evaluation uncertainty of venture capitalists' investment criteria

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ABSTRACT

This paper analyzes the decision process of venture capitalists. The study focuses on aligning the evaluation uncertainty in the decision criteria of venture capitalists with the progress of the process. The reasoning builds from the concept of search, experience and credence qualities, which was developed in the economics of information and allows the identification of the varying uncertainty of a single decision criterion compared to other criteria, along with uncertainty variations throughout the process. Exploratory empirical evidence suggests that in the early steps of the process in particular, management criteria are uncertain, while at the end of the process other criteria couple with uncertainty.

1. Introduction

This research asks how venture capitalists organize their selection and decision processes. Present academic literature on venture capital finance analyzes the pre- and post-contracting phases of the venture capital process in depth (for an extensive review cf. Wright et al., 2003). With respect to the post-contracting phase, scholars discuss such questions as the successful cooperation of venture capitalists and entrepreneurs (Cable and Shane, 1997; Kollmann and Kuckertz, 2006), monitoring of venture capital investments (Kaplan and Strömberg, 2003) and methods of adding value after reaching an investment decision (Sapienza, 1992; Jain, 2001; Chen, 2009). Concerning the pre-contracting phase, several works identify the central steps of the decision process (Wells, 1974; Tyebjee and Bruno, 1984; Fried and Hisrich, 1994), and determine the relative importance of particular investment criteria to a final decision (MacMillan et al., 1985; Muzyka et al., 1996). In addition, Shepherd et al. (2003) and also Franke et al. (2006) detect potential cognitive biases amongst venture capitalists.

This paper extends the first two aspects of extant venture capital theory with respect to the pre-contracting phase, that is, the study integrates a process perspective with the theoretical knowledge of investment criteria applied by venture capitalists. Such an integrative and dynamic perspective can provide a more realistic venture capital theory as well as addressing criticisms that prior studies have merely collected empirical evidence about global investment criteria which is only applicable to the complete investment process (Franke et al. 2006). Since numerous prior studies address the relevance of particular investment criteria, this paper focuses on the specific uncertainty associated with the evaluation of each single criterion and analyzes how this uncertainty varies throughout the investment process. Empirical knowledge about this uncertainty may then be the basis for future instruments and strategies for managing a single investment criterion depending on the progress of the investment process.

This article develops its argument along the following lines. After reviewing extant knowledge about venture capital finance, the paper introduces the economics of information as a theoretical concept that allows the identification of evaluation uncertainty. Subsequently, the paper explores the relationship between venture capital finance and the economics of information. Next, the article presents the methodology and results of an explorative empirical study among 81 venture capitalists from German-speaking Europe. The paper concludes with final remarks that underscore the implications for efficient management of the investment process and also shows how information about evaluation uncertainty theoretically completes the existing knowledge about the relevance of venture capital investment criteria.

2. Theoretical background

2.1. Venture capital investment process

The investment process of venture capital is part of the "venture capital cycle" (Gompers and Lerner, 2000), and has not changed significantly since the first academic studies in the 1970s (Wright et al., 2003, p. XXIV). Over the complete cycle, venture capitalists raise funds (fundraising), invest those funds in an investment process (deal origination, screening, evaluation, structuring), manage their investments once an investment decision has been made (monitoring and value adding), and eventually realize any profits from their investments (exit).

Wells was one of the first to describe the cooperation of the venture capitalist and the entrepreneur in detail (1974). However, scholars have paid more attention to Tyebjee and Bruno (1984) when

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conceptualizing the process. According to them, venture capitalists invest in five distinct steps. In the first step, deal origination, it is the main task to discover promising investment opportunities. In the second step, deal screening, the investor reduces the overabundance of investment opportunities to a manageable quantity. In the third step, deal evaluation, investors carefully analyze the potential portfolio company. In the fourth step, deal structuring, venture capitalist and entrepreneur clarify the terms of the deal between themselves. The fifth step, post-investment activities, combines all the activities of venture capitalists that primarily aim at supporting the company management and preparing future divestment. This early systemization of the process still holds true, however, some authors (Fried and Hisrich, 1994) propose a more sophisticated analysis of the deal evaluation step, stating that it could be differentiated into a cursory evaluation and a more formal due diligence (so-called first-phase and second-phase evaluation).

This paper provides an explanation of the selection and decision mechanisms employed by venture capitalists, and is therefore concerned with all of the above steps except the last (post-investment activities). However, for pragmatic reasons, the study needs to concentrate on the most important steps in the investment process when proceeding towards the empirical analysis. Given that this paper seeks to integrate the investment process, investment criteria and evaluation uncertainty with an empirical study of venture capitalists, the following discussion focuses on the screening, the evaluation, and the structuring phases. It is important to underscore that only pragmatic reasons drive this shortened approach, and the reader should not interpret its use as an alternative conceptualization of the process. Rather, the study needs a sufficient number of process steps to facilitate an empirical process perspective and allow identification of variations in evaluation uncertainty of a single investment criterion — for instance, by comparing early and late phases of the process. Against this background, the question emerges, on what criteria do venture capitalists actually base their decisions about an investment during the process?

2.2. Venture capital investment criteria

Venture capital researchers frequently address questions of venture capitalists' investment behavior, along with due diligence and its related issues, by focusing on venture capitalists' investment criteria. Empirical studies of the decision making behavior of venture capitalists suggest that characteristics of the entrepreneur and the entrepreneurial team are of the utmost importance: The influence of a management team's quality on venture performance is not merely a cliché popular with venture capitalists; it is also supported by various empirical studies. For example, the seminal study by MacMillan et al. (1985) proposes that five of the 10 most important decision criteria are related to the personality or experience of the entrepreneurs. Moreover, those authors emphasize the fact that investment criteria usually fall into five categories: the personality of the entrepreneur, the experience and qualifications of the entrepreneur, the product/service, the market and financial considerations. Muzyka et al. (1996) provide some evidence that European venture capitalists, especially, attach importance to management team criteria rather than the characteristics of the lone entrepreneur. According to these authors, product and market criteria are only of average importance, while criteria of the fund and the respective terms of a deal's structure are of minor importance.

Similar to the pragmatic reduction of process steps introduced above, the overabundance of known investment criteria forces researchers to confine themselves to just the most relevant. Venture capital literature features plenty of possible investment criteria, and due diligence checklists may well include up to 400 different criteria. Unfortunately, it is rarely feasible to use such a comprehensive catalogue of investment criteria in an empirical study; this paper concentrates on 15 important investment criteria, listed in Table 1. Despite the reduced number of criteria, this is a catalogue that a venture capitalist would most likely perceive as largely complete. The study achieves this goal by concentrating on the categories that MacMillan et al. (1985) suggest, assigning three important investment criteria to each category. This assignment was conducted with the explicit goal of achieving the greatest possible approximation to a complete catalogue of investment criteria. It only considered investment criteria proven to be of significance for actual decision making in at least two prior empirical studies. Participants in the pretest (see below) of our empirical study thought this catalogue to be near complete, but did however, opt for the introduction of one additional criterion that reflects the appropriateness of an entrepreneur's character for venture capital as a specific form of entrepreneurial finance, and distinguishes innovative and ambitious entrepreneurs pursuing growth-oriented business models (Morris et al., 2005) from entrepreneurs operating, for instance, mom and pop businesses. The new criterion proved to be of relevance in all pretests.

2.3. Search, experience and credence qualities as a theoretical point of reference

Every decision of a venture capitalist on an investment proposal is primarily a decision about the quality of that proposal. Therefore, further analysis requires a theoretical framework that allows us to classify investment proposals by the various choices made to evaluate quality. However, a direct measurement of the uncertainty and the amount of uncertainty accompanying a proposal's assessment – for example, by utilizing direct questions and Likert-type scales – could be quite problematic. For instance, when questioned directly, survey

Table 1
Venture capitalists' investment criteria.

Factor	Investment criteria	Evidence of criterion's relevance
Personality of the entrepreneur	“VC character”	Pretest
	Leadership capabilities	MacMillan et al. (1985), Robinson (1987)
	Commitment	Dixon (1991), Muzyka et al. (1996)
Experience of the entrepreneur	Track record	Flynn (1991)
	Technical qualification	Shepherd (1999b)Franke et al. (2006)
	Business qualification	Shepherd (1999b)Franke et al. (2006)
Product or service	Innovativeness	MacMillan et al. (1985)Mason and Stark (2002)
	Patentability	Tyebjee and Bruno (1984)MacMillan et al. (1985)
	Unique selling proposition	Mason and Stark (2002)
Market characteristics	Market volume	Tyebjee and Bruno (1984), Mason and Stark (2002)
	Market growth	Tyebjee and Bruno (1984), Mason and Stark (2002)
	Market acceptance	Tyebjee and Bruno (1984), Mason and Stark (2002)
Financial characteristics	Fit to investment strategy	Muzyka et al. (1996), Mason and Stark (2002)
	Return on investment	Tyebjee and Bruno (1984), MacMillan et al. (1985)
	Exit possibilities	Muzyka et al. (1996); Mason and Stark (2002)

participants are very likely to systematically rate their subjective uncertainty as low so as not to appear incompetent. Hence, this study resorts to a concept taken from the economics of information that will be introduced below: the concept of search, experience and credence qualities (SEC qualities).

SEC qualities provide a suitable theoretical framework for the indirect measurement of subjective evaluation uncertainty of different goods (Nelson, 1970; Darby and Karni, 1973). This concept originates in the economics of information (Kleinaltenkamp and Jacob, 2002; Riley, 2001) and has proved valid in various empirical studies (e.g. Brush and Artz, 1999; Miller, 2002), so researchers can rely on already-validated scales. The framework is of practical relevance as well, for instance, in the United States the Federal Trade Commission (FTC) successfully applied the concept to regulate advertising activities (Ford et al., 1988). Following the basic definition of this framework, each and every type of goods may be described by three different qualities (Darby and Karni, 1973, p. 69): "Search qualities which are known before purchase, experience qualities which are known costlessly only after purchase, and credence qualities which are expensive to judge even after purchase." Taken together, every good can be considered to be a combination of these three basic qualities.

As a matter of principle, SEC qualities are applicable to every conceivable economic type of goods, and therefore also to a venture capitalist's investment proposal. For instance, venture capitalists can decide immediately to a large extent whether a venture operates in an industry of interest (search quality); only later will they be able to assess the actual work effort of the entrepreneur (experience quality). On the contrary, the entrepreneur's concrete commitment towards the venture will never be completely recognizable (credence quality).

It is important to note, however, that the attribution of qualities to a given type of goods can never be objective (Ford et al., 1988; Adler, 1996), but rather results from the subjective perspective of the demand-side (in this case, the venture capitalist). Since every type of quality relates to a different degree of ratability, it becomes evident that there is a hierarchy of qualities that depends on the respective level of evaluation uncertainty (Adler, 1996): goods primarily determined by search qualities are associated with only a low degree of evaluation uncertainty, while goods determined primarily by credence qualities are associated with a comparatively high degree of evaluation uncertainty. Experience qualities, however, have an evaluation uncertainty located somewhere in between search and credence qualities.

Earlier empirical studies on SEC qualities tend to collect global judgments about relatively simple goods (for example bicycles or furniture, Nelson, 1970), thereby reducing the complexity of real transaction processes. As a result, the economics of information literature now frequently calls for studies focusing on the relation between SEC qualities and certain attributes of the respective goods to provide a more complete picture (Adler, 1996). In analyzing the investment proposal of a venture capitalist, this approach becomes necessary, because the particular attributes of a potential investment are commensurate with the investment criteria of venture capitalists — all of which carry different levels of uncertainty. Of course, such an approach implies a catalogue of criteria that is as complete as possible and valid for all enquirers. With venture capital financing, the results of several empirical studies logically form such a catalogue for the purposes of the present research project (see Table 1). All in all, SEC qualities are a suitable theoretical concept to identify the evaluation uncertainty not only of a complete investment proposal, but of a particular investment criterion as well. Moreover, given its three-dimensional character, the concept allows more complex analyses in comparison to more traditional approaches. Therefore, this study will apply this concept to venture capitalists' investment criteria in the following section while simultaneously considering the process character of every venture capital investment.

2.4. Venture capital finance from the perspective of search, experience and credence qualities

Earlier studies on the relevance of investment criteria lead to the conclusion that the impact of a single criterion on an investment decision is not stable; rather criteria will change depending on a given market situation or on the level of experience of a given venture capitalist. Furthermore, empirical findings indicate that at least some criteria are determined by the target industry of the venture capitalist; for instance, the scientific reputation of the founding team seems to be of greater importance when financing biotech ventures than it would be in other industries (Champenois et al., 2006).

Similar assumptions can be made about the evaluation uncertainty of investment criteria that are at the very heart of this study, which assumes that different criteria show different degrees of evaluation uncertainty at different points in the investment process. Such differences are caused by

- the subjective assessment of criteria by individual venture capitalists,
- different amounts of resources being allocated to the evaluation of an investment proposal and
- different levels of information that develop in the course of the venture capital process.

For instance, it is definitely easier to ascertain that a certain company is targeting a market that is of interest to the venture capitalist, than it is to assess the potential market acceptance of innovative products for which entrepreneurs seek funding in order to develop a prototype. Equally, it is fair to assume that venture capitalists perceive a higher degree of evaluation uncertainty at the beginning of the process with an unfamiliar venture than at the end of the process. Therefore, venture capitalists' investment criteria carry different levels of evaluation uncertainty. This evaluation uncertainty is reflected through SEC qualities and changes with the progress of the investment process. Fig. 1 illustrates how evaluation uncertainty, SEC qualities and the dynamics of the venture capital process are related to each other.

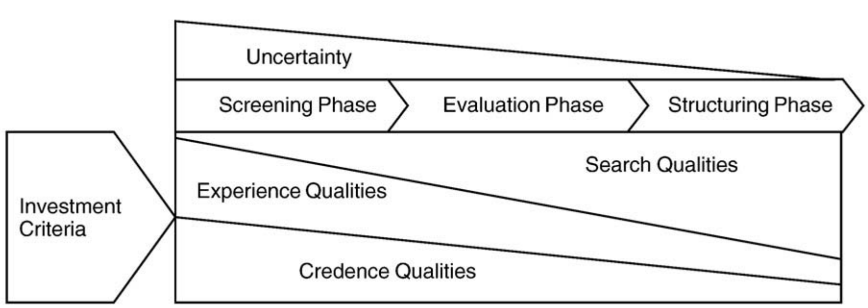

Fig. 1. Search, experience, and credence qualities and the venture capital process.

This paper develops its arguments against the background of research results on one key group of criteria — the management (team) criteria. Several studies have found that venture capitalists ascribe exceptional relevance to this group of criteria. Sayings such as "I invest in people, not in ideas" underscore these findings, which, to a great extent, go back to the significant contribution of the entrepreneur to the potential success of every venture (Stuart and Abetti, 1987). An additional explanation can be found in the difficulty venture capitalists experience in evaluating these criteria. The opportunity costs of the entrepreneur, along with asymmetric information, are possible reasons for the difficulties in evaluating management criteria. Opportunity costs are of relevance for the actual behavior of both entrepreneurs and venture capitalists, and can be the cause of uncooperative behavior on both sides (Cable and Shane, 1997). For entrepreneurs, there is the risk that venture capitalists may terminate the venture because of a more attractive market opportunity. It is equally complicated to decide whether entrepreneurs are willing to carry the risk of a venture because they believe in the success potential of the project, or because they see only very low opportunity costs, for example, for reasons of incompetence and missing qualifications (Gifford, 2003, p. 48).

Asymmetric information is a second reason for possible evaluation uncertainty. As Gompers and Lerner (2000, p. 3) point out, "Venture investors typically concentrate in industries with a great deal of uncertainty, where the information gaps among entrepreneurs and investors are commonplace," and where venture capitalists cannot directly observe the larger part of the entrepreneurs' competencies and motivation. As investors often do not adhere to non-disclosure agreements (Cable and Shane, 1997, p. 149), entrepreneurs may tend to withhold critical information, for example, about specific aspects of the venture's technology; moreover, they may withhold negative information so as not to make funding (or the continuation of the venture) impossible. Similar reasons apply to the danger of entrepreneurs exaggerating the potential performance of the venture. Consequently, this study proposes that management criteria carry excessive evaluation uncertainty, given that appropriate methods to assess the character and competencies of entrepreneurs seeking venture capital still seem to be missing (Smart, 1999). From an economics of information viewpoint, this study therefore concludes that the evaluation uncertainty of management criteria relates to a disproportionate amount of perceived credence qualities.

3. Method

3.1. Research design

The empirical study is based on a standardized written survey instrument. The questionnaire measures a respondent's ability to evaluate an ideal-type investment proposal at various hypothetical points in the investment process. For pragmatic reasons, we refrained from including each and every step of the previously introduced process model, in order to not overload the respondents (completing the whole questionnaire required approximately 30 minutes of an individual respondent's time). Survey participants were asked to recall three different steps of the process — the screening, the evaluation and the structuring phases (Tyebjee and Bruno, 1984). For each stage of the process, survey participants were asked to provide their assessment of the 15 investment criteria that prior studies had proved to be of relevance. SEC qualities were measured based on scales suggested and validated by Ford et al. (1990) and Adler (1996) (see Table 2). Using this approach, it is possible to consider the dynamic component of the investment process and to identify possible changes in perceived evaluation uncertainty between the selected points in time.

Following the development of the research design, the questionnaire was extensively pretested. This pretest was designed as an iterative process; rather than conducting the pretest simultaneously with several test participants, the chosen method was to gather insights from each trial and incorporate them into a new, improved version of the questionnaire to be tested again. In order to shed light on the content reliability of the questionnaire, researchers conducted six pretests with a diverse group of test respondents that included early-stage and expansion-stage investors, as well as independent and captive investors. Each iteration eliminated unsuitable question formats and substituted new formats – after four of six possible iterations, the survey instrument was of sufficient reliability to start data collection.

3.2. Data collection and sample

Venture capital investment managers from German-speaking Europe, that is Germany, Austria, and Switzerland, contributed the data ($n = 81$). At the time of data collection there were approximately 300 venture capital firms which were included in the membership directories of three national professional associations which had recruited nearly all active venture capital firms. These associations are: the German Private Equity and Venture Capital Association (BVK), the Austrian Private Equity and Venture Capital Organization (AVCO) and the Swiss Private Equity and Corporate Finance Association (SECA). The second step was to identify those individuals within each firm who were sufficiently familiar with the venture capital process to supply valid answers. Given that the focus of this study is on individual rather than on organizational decision making, the authors accepted multiple respondents per firm. Every individual received a paper invitation mail and then follow up email reminders at intervals of four weeks. The questionnaire was distributed with a cover letter that explained the purpose of the study and clearly indicated the responsible researchers and their institutional affiliation. Moreover, the cover letter assured respondents that they would remain anonymous and their answers would be treated as confidential. At the end of the data collection phase, 81 analyzable questionnaires had been received (9% response rate), which is in line with similar studies on venture capital finance (for example, Franke et al., 2006; MacMillan et al., 1985; Muzyka et al., 1996). Comparison of early and late respondents by a Mann-Whitney U-test found no indication of non-response bias.

4. Results

4.1. Descriptive analysis

As with any other study, whether or not the collected data represent a valid picture of the underlying population is important. Whether the results are generalizable can at least to some extent be concluded from the descriptive analysis of the available data set. The

Table 2
Operationalization of two exemplary investment criteria at the point of the due diligence.

At the time of the due diligence: To what degree is it possible to you to evaluate certain aspects of a typical investment proposal? Please assign 100% in total to the three categories below				
At the time of the due diligence I am able to assess the quality of the aspect …	… already before the due diligence (up to … %)	… only after the due diligence (up to … %)	… not even after the due diligence (up to … %)	Total
… track record of the entrepreneur				100%
… innovativeness of the product or service				100%

univariate analysis results show that most of the responding venture capitalists had a professional background in management (72.8%) rather than in technology. For a study on the decision behavior of venture capitalists it is vital to seek responses from experienced venture capitalists (McNally, 1994, p. 280). The venture capitalists in our sample were indeed experienced investors: they had a mean professional experience (i.e. work experience post qualification) of 13 years (SD = 8.39), of which they had spent 6.5 years (SD = 4.87) in the venture capital industry; and more than three-quarters were responsible for the total venture capital process. Therefore, on the whole, it is safe to assume the survey participants were sufficiently experienced. Of the respondents, 79% work in German firms, which were, on average, established in 1997; the remaining 21% were based in Austria and Switzerland. These venture capital firms had average funds of €80 million under management, which is comparable to volumes usually reported in the European setting (German Venture Capital Association, 1994–2006). The same applies to the range of investment size: venture capitalists in our sample sought investments of between €1 million and €5 million. The analysis of the preferred target industries revealed that investors are interested in firms operating in the digital economy ("communication technologies" and "computer related" each scored more than 50%), mostly ignoring firms operating in saturated markets sectors like retail. Overall, the data show no sign of bias that would prevent generalizing these findings to German-Speaking Europe.

4.2. Evaluation uncertainty

This section relates the venture capital process and investment criteria by pinpointing the evaluation uncertainty of a single investment criterion depending on the progress of the process. In addition to applying the economics of information to the phenomenon of venture capital finance, the study attempts to explain the evaluation uncertainty of management criteria. Table 3 provides the distribution of SEC qualities of an ideal-type investment proposal, broken down into 15 different investment criteria. The percentages indicate the amount of a particular SEC quality that respondents perceive at a given stage of the venture capital process. For instance, in the screening stage the overall fit of the proposal to the investment strategy of the venture capital firm is characterized by 39.4% search qualities, 50.0% experience qualities and 10.6% credence qualities.

In order to discover the relative uncertainty of a single criterion, the paper compares the respective assessments with the aggregated global assessment, which is obtained by calculating the arithmetic mean across all singular assessments. In this way the study identifies those criteria with distributions of SEC qualities that differ significantly from the global distribution, allowing for more sophisticated conclusions on a criterion's evaluation uncertainty. This permits conclusions about which criteria are, due to their better or worse ratability, related to a lower or higher level of uncertainty. Moreover, by comparing data from the three different process steps, it is possible to identify the varying evaluation uncertainty throughout the process. Towards the end of the process in particular the data are not distributed normally, so the non-parametric Wilcoxon-Test is used to test for significant differences. Table 4 summarizes the results of this test and indicates the deviation of every investment criterion, as a percentage, from the aggregated global assessment of SEC qualities at a given stage of the venture capital process. We suggest that the results should be interpreted in one of two ways:

- Venture capitalists perceive a criterion that shows a significantly higher amount of credence qualities, along with a significantly lower amount of search qualities or experience qualities, as more uncertain than average.
- Venture capitalists perceive a criterion that shows a significantly lower amount of credence qualities, along with a significantly higher amount of search qualities or experience qualities, as less uncertain than average.

That is, for the existence of an uncertain investment criterion, not only must higher credence qualities be perceived at a certain point of the process, but the two remaining SEC qualities need to be perceived to a significantly lower degree. Moreover, to avoid a potential type I error (false positive findings), this study only accepts results that are significant at the 0.001 level.

As a result, two criteria are significantly easier to evaluate than the others in the screening phase. One is the innovativeness of the venture's product/service, which shows approximately 10% fewer credence qualities and more than 6% higher experience qualities compared to the global assessment. That is, venture capitalists are, at the beginning of the investment process, already quite sure that they will be able to decide whether the degree of innovativeness meets their requirements. The other, and even better evaluable, is whether an investment proposal fits in with the venture capital firm's overall investment strategy. The credence qualities of this criterion are about 20% less than average, while the search qualities exceed the average by a similar amount.

These two criteria can be contrasted with four other criteria that are, at the beginning of the investment process, of more than average uncertainty. In particular, the entrepreneur's commitment proves to be problematic, because it is related to more than 14% higher credence qualities. These diverging credence qualities are associated with significantly lower search and credence qualities (-6% and −9%,

Table 3
Relative distribution of search qualities (SQ), experience qualities (EQ) and credence qualities (CQ) of an ideal-typeinvestment proposal — separated into single investment criteria (n = 81).

Factor	Criterion	Screening phase (%)			Evaluation phase (%)			Structuring phase (%)		
		SQ	EQ	CQ	SQ	EQ	CQ	SQ	EQ	CQ
Personality of the entrepreneur	"VC character"	20.6	43.3	36.1	51.1	38.8	8.9	81.7	10.6	7.8
	Leadership capabilities	13.9	48.0	38.0	48.2	41.1	10.8	81.3	9.7	8.7
	Commitment	12.9	40.7	46.4	42.8	41.7	14.8	77.4	12.3	10.4
Experience of the entrepreneur	Track record	20.0	56.0	23.7	52.8	41.0	6.8	86.4	9.6	4.0
	Technical qualification	16.0	50.9	33.1	45.6	44.9	9.4	83.5	9.2	7.3
	Business qualification	15.2	52.8	31.6	51.1	40.6	8.3	82.6	9.9	7.6
Product or service	Innovativeness	21.9	55.7	22.2	51.8	40.2	7.8	83.3	8.5	8.2
	Patentability	12.7	51.5	35.8	43.2	47.9	9.1	85.0	8.8	5.7
	USP	17.2	53.9	28.8	49.2	39.0	11.4	82.1	8.6	9.3
Market characteristics	Market volume	24.3	50.0	25.7	49.6	41.3	8.9	81.8	7.3	9.9
	Market growth	21.0	50.1	29.0	48.3	40.3	11.1	83.2	7.3	10.2
	Market acceptance	13.6	45.5	40.4	37.9	39.4	22.8	69.9	10.7	19.2
Financial characteristics	Fit to investment strategy	39.4	50.0	10.6	70.8	26.4	2.8	91.7	6.5	1.9
	Return on investment	10.2	44.4	45.3	40.3	38.4	20.7	71.4	12.0	16.7
	Exit possibilities	20.9	48.9	30.2	47.9	37.1	15.0	75.8	11.7	12.2
Global assessment		18.7	49.5	31.8	48.7	39.9	11.2	81.2	9.5	9.3

Table 4
Deviation of investment criteria from the overall assessment of an investment proposal's search qualities (SQ), experience qualities (EQ) and credence qualities (CQ) — results of the Wilcoxon Test (n = 81).

Factor	Criterion	Screening phase (%)			Evaluation phase (%)			Structuring phase (%)		
		SQ	EQ	CQ	SQ	EQ	CQ	SQ	EQ	CQ
Personality of the entrepreneur	"VC character"	1.9	−6.1*	4.3	2.4	−0.1	−2.3*	−0.6	1.1	−1.5
	Leadership capabilities	−4.7***	−1.4	6.2***	−0.5	1.1	−0.5	0.2	0.2	−0.6*
	Commitment	−5.8***	−8.8***	14.6***	−5.9**	1.8	3.5*	−3.8*	2.8**	1.2
Experience of the entrepreneur	Track record	1.4	6.5**	−8.1***	4.1	1.1	−4.5***	5.3***	0.1	−5.3***
	Technical qualification	−2.7**	1.5	1.3	−3.1	5.0**	−1.8**	2.4**	−0.3	−2.0***
	Business qualification	−3.5***	3.3**	−0.2	2.4	0.7	−2.9***	1.5	0.3	−1.7**
Product or service	Innovativeness	3.2*	6.2***	−9.6***	3.1	0.2	−3.4***	2.1**	−1.0**	−1.1**
	Patentability	−5.9***	2.0	4.0	−5.5*	7.9***	−2.1***	3.9***	−0.7*	−3.6***
	USP	−1.4	4.5*	−2.9	0.5	−0.9	0.1	0.9	−0.9	0.1
Market characteristics	Market volume	5.6**	0.5	−6.1***	0.9	1.3	−2.3***	0.6**	−2.3***	0.7
	Market growth	2.4	0.7	−2.8*	−0.4	0.4	−0.2	2.1*	−2.2**	1.0
	Market acceptance	−5.1***	−3.9*	8.6***	−10.8***	−0.6	11.5***	−11.3***	1.2	9.9***
Financial characteristics	Fit to investment strategy	20.8***	0.6	−21.3***	22.1***	−13.5***	−8.4***	10.5***	−3.0***	−7.4***
	Return on investment	−8.5***	−5.0	13.5***	−8.4***	−1.5	9.5***	−9.8***	2.5	7.4***
	Exit possibilities	2.3	−0.6	−1.6	−0.8	−2.9*	3.8*	−5.3***	2.2	2.9*
Global assessment		18.7	49.5	31.8	48.7	39.9	11.2	81.1	9.5	9.3

***$p \leq 0.001$, **$p \leq 0.01$, *$p \leq 0.05$. Negative numbers indicate a relatively lower amount of a particular SEC quality compared to the overall assessment at a given stage of the venture capital process, whereas positive numbers indicate a relatively higher amount of a particular SEC quality.

respectively). The entrepreneur's leadership capabilities are almost equally hard to evaluate (−5% search qualities and +6% credence qualities). Market acceptance turns out to be even more difficult with respect to evaluation uncertainty: this criterion is accompanied by approximately 5% lower search qualities and approximately 9% higher credence qualities. Lastly, the assessment of potential return on investment is complicated by more than 13% higher credence qualities.

When progressing to the evaluation phase, this picture changes dramatically. While in earlier phases of the investment process, management criteria (particularly criteria with respect to the personality of the entrepreneur) were especially challenging for venture capitalists, they now adjust to the global assessment. Market acceptance and return on investment continue to be problematic criteria, while the fit to the firm's investment strategy continues to be significantly better evaluable. Only one new criterion turns out to be only marginally more easily evaluable: the patentability of the product/service, with slightly lower credence qualities (−2%) and higher experience qualities (+8%).

After the transfer to the structuring phase, results change only marginally. Investment criteria that had been identified as critical remain uncertain (market acceptance, return on investment) and criteria that were found in the middle of the process to be better to evaluate (patentability, fit to investment strategy) are complemented by only one new criterion being better evaluable than average: the entrepreneur's track record is characterized by about 5% higher search qualities and 5% lower credence qualities. Controlling for the professional and the venture capital experience of survey participants, their position in the venture capital firm (top decision maker vs. investment manager) and the preferred investment industries did not result in significant differences.

5. Conclusion

Studies addressing venture capital decision making – especially studies from the 1980s and early 1990s – have occasionally been criticized. In particular, academics have questioned the validity of studies relying on retrospective questioning, self-completed questionnaires (as compared to observation of authentic decision processes) and direct reporting of decision makers (for an overview cf. Shepherd, 1999a, p. 622). These methods can lead venture capitalists to report how they believe they decide, instead of reporting how they actually reached decisions. When questioned directly, venture capitalists tend to over-stress criteria irrelevant to day-to-day business, while under-stressing significant criteria concerning the profitability and survivability of a given venture. Eventually, researchers have to admit that the analysis of the relevance of a particular decision criterion is only partially pertinent to the successful management of the investment selection process. This is because the factors that make a venture capital investment successful do not necessarily need to be congruent with actual decision criteria. Although there are frequent calls in the academic literature for studies to establish a direct link between investment criteria and success (for example, Tyebjee and Bruno, 1984; Franke et al., 2006), empirical results on this relationship are still missing. Consequently, studies of investment criteria have, up to now, been of more interest to entrepreneurs seeking venture capital funding than they have to venture capitalists looking for appropriate investment criteria. Studying the evaluation uncertainty of investment criteria as the present study does is therefore an important step on the way towards linking a single criterion to its contribution to the success of an investment decision.

However, like all research, this paper is not without limitations. The study's focus on German-speaking Europe presumably prevents generalizing the results to other parts of the world; however, the data at-hand potentially adds to a global perspective on venture capital. Second, the study's focus on the individual level impedes generalizing the results to the organizational level, since larger venture capital firms might be included to a more than proportionate amount in the sample. Last, while our approach allows to identify excessive and moderate levels of uncertainty, it does not allow to identify threshold values above which venture capitalists would feel uncomfortable with a particular investment criterion's uncertainty and below which the level of uncertainty would be perceived as acceptable. Moreover, descriptions of the venture capital process still need to be linked to actual investment performance. Future research efforts could build on this by investigating the dynamics of the venture capital process in research settings that would be even more authentic. Nonetheless, from this paper's empirical findings several implications for actual management result are worth considering. These implications should prove relevant for both entrepreneurs and venture capitalists.

With respect to entrepreneurs seeking venture capital funding, this paper may lead to an improved collaboration between entrepreneurs and venture capitalists. Every criterion utilized in this study was extremely relevant in some prior study. Entrepreneurs can now aim to provide exactly the information at the right point in time that most minimizes the uncertainty of venture capitalists. Such an approach has the potential to simplify capital acquisition. For instance, entrepreneurs should signal their readiness and commitment to the intended venture from the very beginning of the process in the most credible way. Postponing this to a later phase of the process heightens the risk that this part of the process will not be reached, because the negotiations will have already been terminated by the venture capitalist.

For venture capitalists, view the results in conjunction with the extant knowledge about the relevance of criteria (for example, Franke et al., 2006; MacMillan et al., 1985). Venture capitalists should particularly stress those investment criteria that have proved to be of high relevance and, at the same time, of high uncertainty. Above all, management criteria are of utmost importance in view of the fact that they not only influence a venture's future success, but, at the same time, are extremely hard to evaluate. Given these findings, comprehending why this part of the evaluation sometimes follows a purely intuitive manner in practice is difficult.

Future research can build on the combination of investment process and investment criteria. The determination of different degrees of uncertainty of different criteria by means of the SEC qualities turned out to be particularly useful, and a great potential exists for deeper analysis in this direction. For instance, relating strategies to reduce the uncertainty of each criterion would be worthwhile. Within the economics of information, a rich set of tools, such as screening and signaling activities on the demand and supply sides (Riley, 2001), are available to make uncertainty more manageable. Consequently, if venture capital research were to consider a criterion's relevance and its uncertainty at the same time, the research would be one step closer to the development of a theoretically grounded evaluation tool for venture investment.

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